



By Nitesh Singh

Job Ready Free Live Training To Become A

SOC

Analyst L1 For Free

Defronix Cyber Security → <https://defronix.com>



What is DNS Footprinting?

DNS Footprinting =
Collecting information about a target using DNS

No hacking yet. Just gathering intelligence.

Attacker's Goal ?

Collect maximum information without
getting detected

What attackers try to find ?

- Subdomains → `admin.company.com`
- Internal services → `vpn.company.com`
- Mail servers → MX records
- Public IPs
- Infrastructure layout
- DNS Record Enumeration



SOC Indicator :

If you see logs like:

admin.site.com

dev.site.com

vpn.site.com

test.site.com

This is NOT normal user behavior
This is recon / footprinting

If Attacker got successful, they can get :

- All subdomains
- Full DNS structure
- Internal naming patterns

This is basically **complete infrastructure leakage**



SOC Detection (Footprinting)

Key Indicators :

- High number of DNS queries from one IP
- Sequential subdomain queries
- Unusual external IP querying many domains

Example:

Source IP: 45.12.x.x

Queries:
admin.company.com
vpn.company.com
mail.company.com
dev.company.com

Conclusion:

Active reconnaissance (DNS footprinting)

DNS POISONING



What is DNS Poisoning?

DNS Poisoning =
Injecting fake DNS responses

Meaning:

domain → wrong IP

Goal :

- Redirect user to fake site
- Steal credentials
- Deliver malware



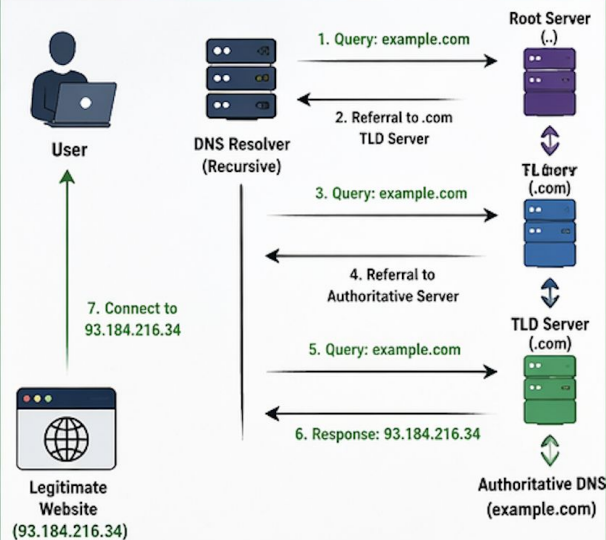
DNS POISONING

HOW IT WORKS, EXAMPLE & DEFENSE



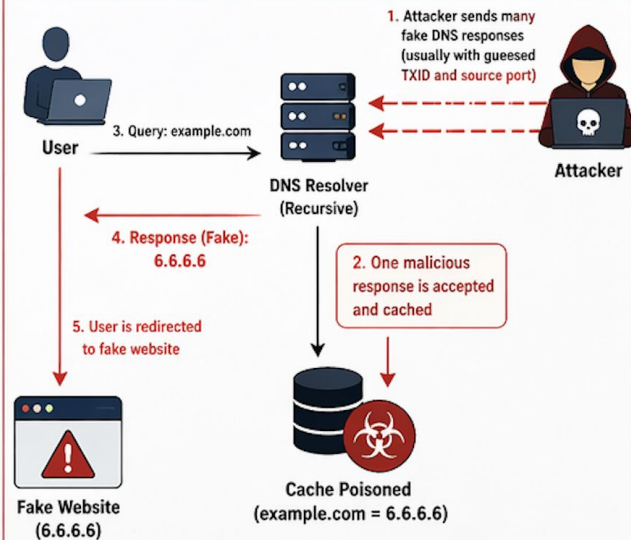
DNS Poisoning is an attack where false DNS records are injected into a DNS resolver's cache, redirecting users to malicious websites.

1. NORMAL DNS RESOLUTION (LEGIT FLOW)



✓ User reaches the legitimate website.

2. DNS POISONING ATTACK FLOW



User is redirected to attacker-controlled site without knowing.

DNS POISONING

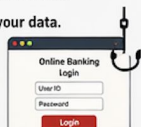


HOW DNS CACHE POISONING WORKS (TECHNICAL VIEW)



EXAMPLE SCENARIO

- 1 You want to visit: www.bank.com
- 2 Attacker poisons DNS resolver with fake entry.
- 3 www.bank.com resolves to 6.6.6.6 (attacker IP)
- 4 You land on a fake bank website that looks identical.
- 5 You enter your credentials.
- 6 Attacker steals your data.



TYPES OF DNS POISONING



CACHE POISONING

Malicious data is injected into DNS resolver cache. Most common and dangerous.



HOSTS FILE POISONING

Attacker modifies the local hosts file to redirect domains to malicious IP.



ROUTER DNS POISONING

Attacker changes DNS settings on the router so all devices use malicious DNS.

INDICATORS OF DNS POISONING (SOC VIEW)



A domain resolves to an unusual or unrecognized IP address.



Sudden change in IP for a domain that normally has a static IP.



Multiple users reporting certificate warnings or website looks different.



High number of failed DNS queries to the same domain.



Unusual external DNS servers communicating with clients.

IMPACT

- 🔒 Credential theft (phishing)
- 🦟 Malware download
- 💰 Financial loss
- 🛡️ Loss of trust and reputation

PREVENTION & DEFENSE

- ✅ Enable DNSSEC to validate DNS responses.
- ✅ Use secure and trusted DNS resolvers.
- ✅ Regularly update DNS software.
- ✅ Implement network monitoring and alerting.
- ✅ Use firewalls/IDS to detect abnormal DNS traffic.
- ✅ Educate users and secure router configurations.



QUICK TIP FOR SOC ANALYST

Always verify:

1. Domain ↔ IP mapping
2. Source of DNS response
3. Frequency and pattern of DNS queries
4. Compare with Threat Intelligence (malicious IP/domain lists)



KEY TAKEAWAY

DNS is the phonebook of the internet. If the phonebook is tampered, you will be directed to the wrong place without even knowing.



Monitor DNS.
Secure DNS.
Stay Ahead.



Real Attack Flow :

1. Attacker poisons DNS
2. User types bank.com
3. DNS returns attacker IP
4. Fake login page shown
5. User enters credentials
6. Credentials stolen

SOC Indicators (DNS Poisoning) :

- DNS Resolution Mismatch
- Sudden DNS Change
- Certificate Warnings
- Multiple Users Affected



DNS FOOTPRINTING vs DNS POISONING

Real SOC Scenario

User complaints:
"Google looks different"

DNS logs:
google.com → unknown IP

Conclusion:
DNS poisoning attack

Feature	Footprinting	Poisoning
Stage	Recon	Attack
Goal	Info gathering	Redirection
Visibility	Low	Medium
Impact	Planning	Compromise



SOC ANALYST MINDSET

When you see DNS logs, don't just ask:

“What domain is this?”

Ask:

- Why is this domain being queried?
- Is the pattern normal?
- Is the response correct?



APT (Advanced Persistent Threat) is:

A **targeted, long-term cyber attack**,
Where attackers **gain access and stay hidden inside a network for a long time**

Advanced

- Uses sophisticated techniques
- Custom malware, zero-days, evasion

Persistent

- Stays in system for **weeks / months / years**
- Maintains access even after detection attempts

Threat

- Organized attackers (often nation-state or skilled groups)

How Advanced Persistent Threat (APT) works



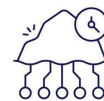
Introduce Malware



Gain Access



Administration Rights



Dig Deeper



Hacker Remain in Place

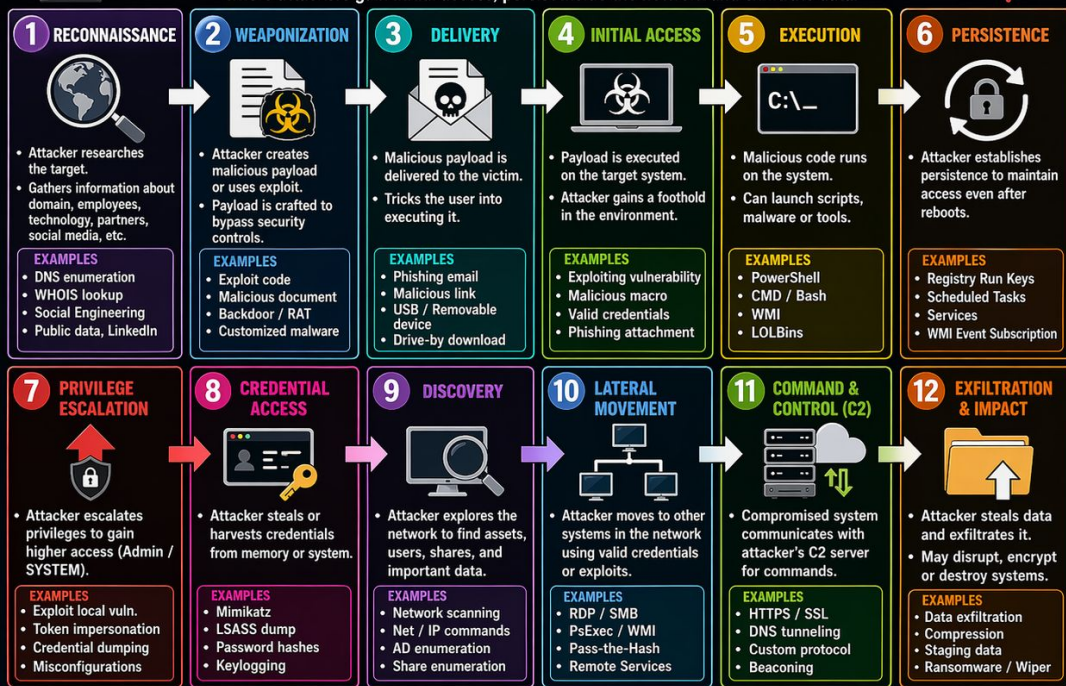


FULL ATTACK CHAIN OF AN APT ATTACK

APT (Advanced Persistent Threat) attacks are targeted, long-term and multi-stage attacks where attackers gain initial access, persist inside the network and exfiltrate data.



Defronix Cyber Security → <https://defronix.com>



APT (Advanced Persistent Threat)
Attack Cycle

APT ATTACKER GOALS

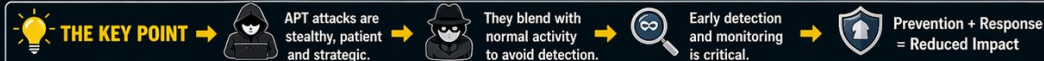
- Steal sensitive data
- Espionage / Intelligence
- Financial gain
- Disruption / Sabotage

COMMON INDICATORS ('IOC VIEW)

- Unusual outbound connections
- Unknown domains / IPs
- Multiple systems accessed
- PowerShell / WMI abuse
- Credential dumping activities
- Lateral movement patterns
- Unusual logon times
- Large data transfers
- Disabled security tools
- Persistence mechanisms

DEFENSE BEST PRACTICES

- Keep systems & software updated
- Use strong email filtering
- Monitor logs & network traffic
- Implement EDR / XDR solutions
- Enforce least privilege access
- Segment network
- Regular security awareness training
- Have an incident response plan





SOC Indicators :

- Low and slow activity
- Unusual login patterns
- Internal lateral movement
- Suspicious PowerShell usage
- Data exfiltration
- Unknown domains (C2)

Real SOC Scenario :

User opens phishing email
↓
PowerShell executed
↓
Mimikatz dumps credentials
↓
Admin login detected
↓
RDP to multiple servers
↓
Data transfer to external IP

This is not random
This is APT-style attack chain



APT vs Normal Attack

Feature	Normal Attack	APT
Duration	Short	Long
Goal	Quick gain	Strategic
Detection	Easier	Hard
Behavior	Noisy	Stealthy

SOC Analyst Mindset

When investigating, Don't think: "This is one alert"

Think: Is this part of a bigger attack chain?